Alessandro Perini

PhD student in Industrial Engineering, University of Rome Tor Vergata

Education

2024–Present	**University of Rome Tor Vergata**, PhD in Industrial Engineering Innovative Doctoral Fund DM 630, funded by the Ministry of University and Research, NextGenerationEU (EU), and Sensor Medica (Italy). "*Development of wearable robotic systems*." Supervisor: Dr. Matteo Russo
2022–2024	**University of Rome Tor Vergata,** MSc in Mechatronics Engineering, cum laude Thesis: "*Development of a tendon-driven hand exoskeleton.*"
2019–2022	**Roma Tre University,** BSc in Mechanical Engineering Thesis: "*Analytical and numerical study of the optimal placement of piezoelectric actuators on tapered beams.*"

Honours & Awards

2026	**Robotics Travel Award,** winner of the Travel Award issued by MDPI Robotics
2024	**Rome Technopole Award (NextGenerationEU):** "*Awarded for the 'ECS000024 Rome Technopole Project' (CUP B83D21014170006)*."
2022–2023	**Scholarship (Youth Fund for Year 2023)** given by the University of Rome Tor Vergata: "*Awardee for merit-based incentives aimed at first-year students who are enrolled in the 2022/2023 academic year*."
2021–2022	**Scholarship (Academic Achievement)** given by Roma Tre University: "*To students who have attained optimal outcomes related to the alignment of the study program and the grades achieved*."

Patents

2025	A. Perini, M. Russo, "Esoscheletro per mano e relativo metodo di fabbricazione" ("Hand exoskeleton and related fabrication method"), Italian Patent IT 102025000009261, filed 24/04/2025.

Technical Skills

Software & Hardware	MATLAB & Simulink, Autodesk Inventor, Fusion 360, ANSYS Fluent, COMSOL Multiphysics, Wolfram Mathematica, LaTeX, MS Word, MS Excel, MS PowerPoint, Affinity Designer, C, C++, 3D printing, machine sewing.
Languages	Italian: native speaker English: full professional proficiency (B2/C1)

Experience

- **PhD R&D Mechanical Design Intern |** Sensor Medica, Italy (Sept 2026 – Present). Spearheading the mechanical design of a robotic gripper and driving the automation of a complete robotic cell to optimize production workflows.
- **Visiting PhD period (six months) |** Technical University of Munich (TUM), Munich Institute of Robotics and Machine Intelligence (MIRMI), ARIES Lab.
- **Admission Committee Assistant |** University of Rome Tor Vergata. Evaluated and scored entrance examinations for the Engineering Science program, ensuring compliance with academic standards.
- **Organizing Committee, PhD EXPO |** University of Rome Tor Vergata. Session chair for the 3 Minute Thesis (3MT) pitch.
- **Student Representative, IEEE Robotics & Automation Society (RAS) |** Technical Committee on Mechanisms and Design, 2026–2027 term.
- **Selected Entry, 2025 James Dyson Award |** "TEXO: a lightweight, accessible glove exoskeleton, fully made of

sustainable textiles."

Media & Outreach

- **Futuro24, Rai News 24 |** Feature interview, broadcast on Rai News 24 (03/11/2026). Available online: rainews.it
- **Casa Italia, Rai Italia |** Short interview, broadcast on Rai Italia (26/10/2026). Available online: raiplay.it
- **Exhibition, Maker Faire Roma 2026 |** "TEXO: Exoskeletons and frontier robotics," Rome, Italy (23–25/10/2026).
- **TG24, Rai News 24 |** Short interview, broadcast on Rai News 24 (23/10/2026).
- ***Il Messaggero* |** Mention in "Maker Faire a Roma, la fiera hi-tech spalanca le porte del futuro," print edition (23/10/2026). Available online: Italian, English.
- ***La Repubblica* |** Full-page article "TEXO, quando la sartoria incontra la robotica," print edition (23/10/2026).

Publications

- **Perini, A.**, Piratoni, M., Tricomi, E., Russo, M., & Masia, L. (2026). Barometric fusion for continuous variable assistance in a soft knee exoskeleton. *IEEE Robotics and Automation Letters* (under review).
- **Perini, A.**, Russo, M., Hasan, M. I., Secciani, N., Ridolfi, A., & Dragusanu, M. (2026). Towards reproducible metrics and benchmarks for soft hand exoskeletons: an open benchmark for grasping characterization. *Advanced Intelligent Systems*, Wiley (under review).
- **Perini, A.**, & Russo, M. (2026). From parametric patterns to tailored textile gloves for soft tendon-driven hand exoskeletons. *IEEE Transactions on Robotics* (submitted).
- **Perini, A.**, & Russo, M. (2026). Design of a rod-driven soft enveloping gripper. *6th IFToMM ITALY Conference*. DOI: 10.1007/978-3-032-35974-2_6
- Beyki, N., **Perini, A.**, Molinaro, L., & Russo, M. (2026). Muscle activation detection with surface electromyography: a combined signal processing and machine learning approach. *International Journal of Advanced Robotic Systems*. DOI: 10.1177/17298806261446421
- Scuderoni, H., **Perini, A.**, & Russo, M. (2026). Design of a single-input, five-output differential actuation unit for underactuated hands. *2026 IEEE International Conference on Robotics & Automation (ICRA 2026)* (accepted).
- **Perini, A.**, & Russo, M. (2026). Underactuation in tendon-driven hand exoskeletons. In *International Conference on Motion Assisting Devices for Limbs* (pp. 230–244). Springer, Cham. DOI: 10.1007/978-3-032-15271-8_13
- **Perini, A.**, & Russo, M. (2025, April). Design of an adaptive tendon-driven hand exoskeleton for stroke rehabilitation. In *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)* (pp. 1–8). IEEE. DOI: 10.1109/RoboSoft63089.2025.11020897

Workshops & Abstracts

- Piratoni, M., **Perini, A.**, Tricomi, E., Patanè, F., & Masia, L. (2026). A soft cable-driven knee-extension exoskeleton for industrial lifting: Design and control strategy. *2026 IEEE International Conference on Cyborg and Bionic Systems (CBS 2026)*.
- **Perini, A.**, & Russo, M. (2026, July). "Bridging the adoption gap: A lightweight textile hand exoskeleton for real-world assistance." In *The Mechatronic Arc: Bridging the Gap Between Physical Design and Computational Intelligence* workshop, AIM 2026, July 7, Genova, Italy.
- **Perini, A.**, & Russo, M. (2026, July). "TEXO: A tendon-driven textile glove exoskeleton." In *HANDS ON! Bringing Upper-Limb Exoskeletons into the Real World* workshop, AIM 2026, July 7, Genova, Italy.
- **Perini, A.**, & Russo, M. (2026, June). "Tailored textile patterns for soft wearable robots." In *SOFT Frontier 3: Soft or Rigid? New Trends in Wearable Robotics* workshop, Vienna, Austria.
- **Perini, A.**, & Russo, M. (2025, April). "Design of a textile-based soft hand exoskeleton." In *Smart Textiles in Soft Robotics* workshop, Lausanne, Switzerland.

Review Work

IEEE Robotics and Automation Letters (RA-L); 2026 IEEE International Conference on Robotics & Automation (ICRA 2026); IEEE RAS/EMBS 11th International Conference on Biomedical Robotics and Biomechatronics (BioRob 2026); Springer Nature, Mechanisms and Machine Science (EUCOMES 2026).

Supervision

- Nifoofar Beyki (2025): Development of sensors for sports biomechanics (with Sensor Medica S.r.l.)
- Matteo Trigona (2026): Progettazione e validazione sperimentale di robot mobile per ispezione
- Hugo Scuderoni (2025, BSc, University of Toulon, SeaTech, France)